

An improved lower bound for the minimum-autocorrelation constant

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Abstract

We construct nonnegative bounded compactly supported functions on the real line that give the lower bound $\mathcal{A} \geq 0.4103779774321214877852987\dots$ for the minimum-autocorrelation constant. This improves the explicitly certified bound $2378625/5958277 = 0.39921356\dots$ reported by Russell. The construction combines a point mass with a monotone degree-ten polynomial density; a one-sided atom-removal argument then produces ordinary functions. Its numerical content reduces to exact rational polynomial identities and 20 Bernstein subinterval certificates. The finite certificate, actual overlap integrals, admissibility, mass formula, and ordinary-function witness bounds have been formalized in Lean 4.19.0. The passage to the supremum constant is given separately as a mathematical argument. As applications, the same construction yields an upper bound below 1.5610183 for the large-multiplicity generalized-difference-basis coefficient and a feasible squared mass below 2.436778 for a continuous sparse-ruler relaxation. An appendix quantifies the loss under independent sampling followed by completion. No improved bound for the complete sparse ruler constant, or global optimality of the construction, is asserted.

1 Introduction and main result

For a nonnegative function $f \in L^1(\mathbb{R})$ of positive mass, put

$$a_f(t) = \int_{\mathbb{R}} f(x)f(x+t) dx, \quad \mathcal{A} = \sup_{\substack{f \geq 0, f \in L^1(\mathbb{R}) \\ \int f > 0}} \frac{\inf_{0 \leq t \leq 1} a_f(t)}{\left(\int_{\mathbb{R}} f\right)^2}. \quad (1.1)$$

For the full L^1 class we interpret the nonnegative integral defining a_f as an extended Lebesgue integral. Tonelli's theorem gives $\int_{\mathbb{R}} a_f = (\int f)^2$, so the infimum in (1.1) is finite. Every witness used below is bounded and compactly supported, and has a finite, ordinary real-valued autocorrelation at every shift. All stated lower bounds hold at every $t \in [0, 1]$, not merely almost everywhere.

The problem asks how large the least autocorrelation on a unit interval of shifts can be, relative to squared mass. It is connected with difference bases in additive combinatorics. Barnard and Steinerberger [1, Theorem 2] established a lower bound greater than 0.37 and the Fourier upper bound

$$\mathcal{A} \leq \frac{1}{K} = 0.410767488189405\dots, \quad K = 2 - 2 \inf_{\xi \in \mathbb{R}} \frac{\sin \xi}{\xi} = 2.434467256422443\dots, \quad (1.2)$$

where the quotient at zero is interpreted by continuity. Madrid and Ramos [2] studied related extremal questions and positive-measure formulations; their strict improvement of this upper bound

in Theorem 1.7 concerns a class with an additional tail-domination condition. We do not use an unconditional numerical improvement from that result. The same minimum-autocorrelation problem appears as Problem 6.6 of Georgiev, Gómez-Serrano, Tao, and Wagner [4]; it is different from the minimum-overlap problem listed there as Problem 6.5.

A public exact-rational certificate of Russell [3] gives

$$\mathcal{A} \geq \frac{2378625}{5958277} = 0.39921356\dots \quad (1.3)$$

Our main contribution is a stronger explicit construction with a short finite certificate and a formalized ordinary-function proof. The comparison in (1.3) is with the specified public certificate, not an assertion of an exhaustive bibliographic priority search.

Define the following exact positive integers:

$$\begin{aligned} N &= 4713977940944586245580976375978623597750000000000000000, \\ D &= 11486917427785951575611071745219515905862879073559791929, \\ D_F &= 11486917430134691792906455293684961315501879073559791929. \end{aligned} \quad (1.4)$$

Set $R = N/D$ and $R_F = N/D_F$.

Theorem 1.1 (Improved minimum-autocorrelation bound). *There is an explicitly specified nonnegative bounded compactly supported function $F_{10^{-10}}$ such that*

$$\frac{a_{F_{10^{-10}}}(t)}{(\int F_{10^{-10}})^2} \geq R_F = 0.4103779773482111379678143\dots \quad (0 \leq t \leq 1). \quad (1.5)$$

For every real $q < R$, there is such an ordinary function f for which

$$\frac{a_f(t)}{(\int f)^2} \geq q \quad (0 \leq t \leq 1). \quad (1.6)$$

Consequently,

$$\boxed{\mathcal{A} \geq R = 0.4103779774321214877852987\dots > 0.4103779.} \quad (1.7)$$

The functions are given in (4.1). The proof separates two issues: an exact polynomial certificate for a measure with an atom, and a monotone extension that removes the atom without losing the required correlation lower bound. The number R is a limiting guarantee for a family of functions; neither attainment of R nor optimality of the polynomial profile is claimed.

Section 2 gives all exact parameters and the overlap reduction. Section 3 describes the finite certificate, and Section 4 proves Theorem 1.1. Section 5 records two applications of the same construction. Section 6 specifies the formalized statements and their reproducibility. Appendix A treats an independent-sampling-and-completion procedure; it is secondary to the continuous lower bound and is not part of the Lean formalization.

2 An exact atomic–polynomial profile

Every terminating decimal specifying a parameter below denotes an exact rational number. Let

$$\begin{aligned} a &= 0.626600210121, & b &= 0.160189995012, & \gamma &= 0.99999999, \\ c_0 &= 1 - b = 0.839810004988, & w &= 1 - 2b = 0.679620009976. \end{aligned} \quad (2.1)$$

Thus $0 < b < w < c_0 < 1$. With the first-kind Chebyshev polynomials defined by $T_0(z) = 1$, $T_1(z) = z$, and $T_{j+1}(z) = 2zT_j(z) - T_{j-1}(z)$, put

$$P(s) = 2s - s^2 + s(1-s)^2 \sum_{j=0}^7 q_j T_j(2s-1), \quad (2.2)$$

where the coefficients are listed in Table 1. The parameterization enforces $P(0) = 0$, $P(1) = 1$, and

Table 1: Exact coefficients in the degree-ten profile.

j	q_j	j	q_j
0	-0.423577846263	4	0.000203543244
1	0.290387575223	5	0.000666727048
2	0.011765259840	6	0.000240602158
3	-0.005796017613	7	0.000112563884

$P'(1) = 0$. The certificate in Section 3 proves $P' \geq 0$ on $[0, 1]$, and hence $0 \leq P \leq 1$ there.

Define

$$\phi(x) = \begin{cases} P((x-b)/w), & b \leq x \leq c_0, \\ 1, & c_0 < x \leq 1, \\ 0, & \text{otherwise,} \end{cases} \quad h(x) = \frac{\phi(x)}{a}, \quad \mu = a\delta_0 + h(x) dx. \quad (2.3)$$

Here δ_0 denotes unit point mass at the origin. The positive measure μ is supported on $[0, 1]$, but is not itself an admissible function in (1.1).

For $0 < t < 1$, the absolutely continuous part of $\mu * \tilde{\mu}$, where $\tilde{\mu}(E) = \mu(-E)$, has density

$$g(t) = ah(t) + \int_{\mathbb{R}} h(x)h(x+t) dx = \phi(t) + \frac{B(t)}{a^2}, \quad B(t) = \int_{\mathbb{R}} \phi(x)\phi(x+t) dx. \quad (2.4)$$

There is also the nonnegative atom $a^2\delta_0$. We extend the displayed expression for g continuously from $(0, 1)$ to its endpoints; more explicitly, the right side of (2.4), with the prescribed values of ϕ , defines g on the whole closed interval. The atom and the density play complementary roles: at small shifts density–density overlap contributes, while at larger shifts the atom–density term becomes substantial.

To compute B , let

$$\begin{aligned} I(z) &= \int_0^z P(s) ds, & u(t) &= 1 - \frac{t}{w}, & v(t) &= \frac{c_0 - t}{w}, \\ H(t) &= w \int_0^{u(t)} P(s)P(s+t/w) ds. \end{aligned} \quad (2.5)$$

The function H is a polynomial in t , although its interpretation as a ramp–ramp overlap is used only for $0 \leq t \leq w$.

Lemma 2.1 (Overlap identity). *For $0 \leq t \leq 1$,*

$$B(t) = \begin{cases} H(t) + w(I(1) - I(u(t))) + b - t, & 0 \leq t \leq b, \\ H(t) + w(I(v(t)) - I(u(t))), & b \leq t \leq w, \\ wI(v(t)), & w \leq t \leq c_0, \\ 0, & c_0 \leq t \leq 1. \end{cases} \quad (2.6)$$

Proof. Split the product into ramp–ramp, ramp–plateau, and plateau–plateau terms. For $0 \leq t \leq w$, the ramp–ramp variable ranges over $b \leq x \leq c_0 - t$. Substituting $x = b + ws$ gives $H(t)$. It contributes zero for larger t .

The ramp–plateau term has $\max(b, c_0 - t) \leq x \leq \min(c_0, 1 - t)$ when this interval is nonempty. Its integral is respectively $w(I(1) - I(u(t)))$, $w(I(v(t)) - I(u(t)))$, and $wI(v(t))$ in the first three intervals of (2.6), and zero in the fourth. A plateau point cannot precede a ramp point when $t \geq 0$. The remaining plateau–plateau term has length $(b - t)_+$. Summing gives (2.6); changes at isolated endpoints do not change the integrals. $\square$

Combining (2.6) with (2.4) defines four rational polynomials g_0, g_1, g_2, g_3 on those same intervals: the added term $\phi(t)$ is zero on $[0, b]$, is $P((t - b)/w)$ on $[b, c_0]$, and is one on $[c_0, 1]$. The formulas agree at all internal breakpoints. Their degrees are at most 21, 21, 11, 0 respectively.

3 The finite rational certificate

Lemma 3.1 (Certified inequalities). *For the exact parameters (2.1) and Table 1,*

$$P'(s) \geq 0 \quad (0 \leq s \leq 1), \quad g_i(t) \geq \gamma \quad (t \text{ in the interval of } g_i).$$

In particular $g(t) \geq \gamma$ *on* $[0, 1]$.

Computer-assisted proof. For a rational polynomial p of degree at most d and a rational interval $[\ell, r]$, write

$$p(\ell + (r - \ell)s) = \sum_{k=0}^d \beta_k \binom{d}{k} s^k (1 - s)^{d-k}. \quad (3.1)$$

Nonnegative coefficients β_k certify nonnegativity on the entire interval. If p_j are the ordinary power coefficients after the affine substitution, the exact conversion is

$$\beta_k = \sum_{j=0}^k p_j \frac{\binom{k}{j}}{\binom{d}{j}}. \quad (3.2)$$

When the coefficient test does not succeed on an interval, halve it and repeat, using rational de Casteljau subdivision. The resulting successful partition is summarized in Table 2. All coefficients of

Table 2: The exact Bernstein certificate. Depth is relative to the indicated initial interval.

Polynomial and initial interval	Degree	Accepted leaves	Maximum depth
P' on $[0, 1]$	9	1	0
$g_0 - \gamma$ on $[0, b]$	21	1	0
$g_1 - \gamma$ on $[b, w]$	21	13	6
$g_2 - \gamma$ on $[w, c_0]$	11	4	3
$g_3 - \gamma$ on $[c_0, 1]$	0	1	0

all 20 accepted leaves are nonnegative. Interval coverage and the polynomial identities are checked exactly. The ancillary file `certificate.json` specifies the input, and `verify.py` reconstructs the polynomials and the partition using only rational arithmetic from the Python standard library. The Lean sources independently check identities, nonnegativity, and coverage; after incorporating the binomial factors and clearing positive denominators, their leaf certificates contain 367 nonnegative

integer coefficients. Thus no floating-point tolerance or grid-sampling assumption occurs in the certificate. $\square$

As an additional algebraic audit, `independent_audit.py` reconstructs all actual overlap integrals from their pairwise support intersections and compares the rational polynomial coefficients with (2.6). It then uses exact Sturm root counts instead of Bernstein positivity: each $g_i - \gamma$ has positive endpoint values and no root in its interval. For monotonicity, it writes $P'(s) = (1 - s)Q(s)$ and checks the corresponding strict positivity of Q on $[0, 1]$. This audit supplies a separate computational check, not a hypothesis in the Lean proofs.

The exact parameters can be checked without rerunning any numerical search. The certificate proves feasibility of this particular profile, not optimality among all profiles or all nonnegative functions.

4 Ordinary functions and the improved bound

For every $\delta > 0$, extend the right-hand plateau by δ and define

$$h_\delta(x) = \begin{cases} h(x), & 0 \leq x \leq 1, \\ 1/a, & 1 < x \leq 1 + \delta, \\ 0, & \text{otherwise,} \end{cases} \quad F_\delta(x) = \frac{a}{\delta} \mathbf{1}_{[0, \delta]}(x) + h_\delta(x). \quad (4.1)$$

The extension of the plateau is part of the construction and cannot simply be omitted.

Lemma 4.1 (One-sided atom removal). *For every $\delta > 0$, the function F_δ is nonnegative, bounded, and supported in $[0, 1 + \delta]$. It belongs to $L^1(\mathbb{R}) \cap L^2(\mathbb{R})$, every shifted product $F_\delta(x)F_\delta(x + t)$ is integrable, and*

$$a_{F_\delta}(t) \geq \gamma \quad (0 \leq t \leq 1). \quad (4.2)$$

Proof. Monotonicity of P gives $0 \leq h_\delta \leq 1/a$, so $0 \leq F_\delta \leq a/\delta + 1/a$. The support and integrability assertions follow. The function h_δ is nondecreasing on $[0, 1 + \delta]$. If $t \in [0, 1]$ and $x \in [0, \delta]$, then

$$0 \leq t \leq x + t \leq 1 + \delta, \quad h_\delta(x + t) \geq h_\delta(t) = h(t).$$

Writing $r_\delta = (a/\delta)\mathbf{1}_{[0, \delta]}$, one cross term in the autocorrelation therefore satisfies

$$\int_{\mathbb{R}} r_\delta(x) h_\delta(x + t) dx = \frac{a}{\delta} \int_0^\delta h_\delta(x + t) dx \geq ah(t). \quad (4.3)$$

Also $h_\delta \geq h \geq 0$ pointwise, whence

$$\int h_\delta(x) h_\delta(x + t) dx \geq \int h(x) h(x + t) dx.$$

The other cross term and the rectangle's self-correlation are nonnegative. Adding these four terms and using Lemma 3.1 gives (4.2). This argument works at $t = 0$ and $t = 1$ directly. $\square$

The mass of the intermediate measure is the exact rational number

$$m = a + \frac{wI(1) + b}{a} = 1.561018228470523 \dots \quad (4.4)$$

Integrating (4.1) gives

$$\int_{\mathbb{R}} F_{\delta} = m + \frac{\delta}{a} > 0, \quad \frac{a_{F_{\delta}}(t)}{(\int F_{\delta})^2} \geq \frac{\gamma}{(m + \delta/a)^2} \quad (0 \leq t \leq 1). \quad (4.5)$$

The exact arithmetic certificate verifies

$$\frac{\gamma}{m^2} = \frac{N}{D}, \quad \frac{\gamma}{(m + 10^{-10}/a)^2} = \frac{N}{D_F}, \quad \frac{N}{D_F} > \frac{4103779}{10^7}. \quad (4.6)$$

Proof of Theorem 1.1. Set $\delta = 10^{-10}$ in (4.5) and use (4.6) to obtain (1.5). As $\delta \downarrow 0$, the right side of (4.5) tends to $R = \gamma/m^2$. Hence for each real $q < R$, some positive δ gives (1.6). Each such F_{δ} is an admissible ordinary function in (1.1), and taking the infimum over t preserves its lower bound. Thus $\mathcal{A} \geq q$ for every $q < R$, which implies $\mathcal{A} \geq R$. $\square$

No equality between a supremum over all of L^1 and a supremum over a restricted function class is required here: it is enough that the constructed witnesses belong to the original class. In particular, the limiting argument does not insert the atomic measure μ into that class.

5 Applications of the same construction

5.1 Generalized difference bases at large multiplicity

For positive integers g, n , let $\eta_g(n)$ be the least size of a finite set $S \subset \mathbb{Z}$ for which every $1 \leq d \leq n$ has at least g ordered representations $d = s - s'$ with $s, s' \in S$. The support of S is not restricted to $[0, n]$. Define

$$\tau = \inf \left\{ \int_{\mathbb{R}} f : f \geq 0, f \in L^1(\mathbb{R}), a_f(t) \geq 1 \text{ for all } 0 \leq t \leq 1 \right\}. \quad (5.1)$$

Kravitz [5, Theorem 1.2] established the discrete–continuous correspondence

$$\lim_{g \rightarrow \infty} \liminf_{n \rightarrow \infty} \frac{\eta_g(n)}{\sqrt{gn}} = \tau = \lim_{g \rightarrow \infty} \limsup_{n \rightarrow \infty} \frac{\eta_g(n)}{\sqrt{gn}}. \quad (5.2)$$

Symmetry of autocorrelation makes the requirement on $[0, 1]$ equivalent to the one on $[-1, 1]$ in that correspondence.

Corollary 5.1. *The coefficient in (5.2) satisfies*

$$\tau \leq \frac{m}{\sqrt{\gamma}} = \sqrt{\frac{D}{N}} = 1.5610182362756142228486109 \dots < 1.5610183. \quad (5.3)$$

Proof. The ordinary function $F_{\delta}/\sqrt{\gamma}$ is admissible in (5.1) by Lemma 4.1. Its mass is $(m + \delta/a)/\sqrt{\gamma}$. Take the infimum over positive δ and use (4.6). The last strict inequality can be checked by squaring positive rational quantities. $\square$

The order of limits in (5.2) is essential. This corollary is not a statement for $g = 1$, nor for sets constrained to $[0, n]$; with the latter restriction, the endpoint distance n has at most one representation.

5.2 A feasible point for the continuous sparse-ruler relaxation

Let $M(L)$ be the least cardinality of a complete sparse ruler $S \subseteq \{0, \dots, L\}$, meaning that $\{1, \dots, L\} \subseteq S - S$. Define

$$B_* = \inf \left\{ \nu([0, 1])^2 : \begin{array}{l} \nu \text{ is a finite positive measure, } \text{supp } \nu \subseteq [0, 1], \\ \nu * \tilde{\nu} \geq \mathbf{1}_{[-1, 1]}(t) dt \end{array} \right\}. \quad (5.4)$$

Domination here is an inequality of measures, not an endpointwise inequality between densities.

Proposition 5.2. *The relaxation satisfies*

$$\liminf_{L \rightarrow \infty} \frac{M(L)^2}{L} \geq B_* \leq \frac{m^2}{\gamma} = \frac{D}{N} = 2.4367779339850293520112621 \dots < 2.436778. \quad (5.5)$$

The two inequalities in (5.5) do not compare $M(L)^2/L$ with D/N .

Proof. Choose any sequence of complete rulers S_L along which $|S_L|/\sqrt{L}$ tends to a finite value s , and set

$$\nu_L = \frac{1}{\sqrt{L}} \sum_{x \in S_L} \delta_{x/L}.$$

The masses are bounded and the supports lie in $[0, 1]$, so a subsequence converges weakly to a finite positive measure ν on that interval, with mass s . Completeness gives

$$\nu_L * \tilde{\nu}_L \geq \frac{1}{L} \sum_{\substack{-L \leq d \leq L \\ d \neq 0}} \delta_{d/L}. \quad (5.6)$$

The right side tends weakly to Lebesgue measure on $[-1, 1]$. On the compact square $[0, 1]^2$, weak convergence of the product measures follows first for functions of the form $u(x)v(y)$ and then for all continuous functions by uniform approximation and bounded masses. Applying this to continuous functions of $x - y$ gives weak convergence of the left side. Testing (5.6) against nonnegative continuous functions proves that ν is feasible in (5.4). Therefore $s^2 \geq B_*$. Taking a subsequence approaching the lower limit proves the first inequality; an infinite lower limit is harmless.

For the second inequality, Lemma 3.1, symmetry, and the nonnegative atom imply $\mu * \tilde{\mu} \geq \gamma \mathbf{1}_{[-1, 1]} dt$. Consequently $\nu = \mu/\sqrt{\gamma}$ is feasible, with squared mass $m^2/\gamma = D/N$. $\square$

For context, the same classical Fourier argument underlying (1.2) gives $B_* \geq K$. Indeed, for a feasible ν let $\rho = \nu * \tilde{\nu} - \mathbf{1}_{[-1, 1]} dt \geq 0$. Using $\hat{\nu}(\xi) = \int e^{-i\xi x} d\nu(x)$ and taking real parts,

$$0 \leq |\hat{\nu}(\xi)|^2 = 2 \frac{\sin \xi}{\xi} + \text{Re } \hat{\rho}(\xi) \leq 2 \frac{\sin \xi}{\xi} + \nu(\mathbb{R})^2 - 2.$$

Optimize in ξ to obtain the bound. Thus the present certificate places this particular relaxation in the narrow interval $[K, D/N]$.

The classical Wichmann construction gives an asymptotic upper coefficient $\sqrt{3}$ for complete sparse rulers; see Saarela and Vanhatalo [6]. Our feasible measure shows that the conditions in (5.4) alone cannot prove $B_* \geq 2.5$ or $B_* \geq 3$. It does not give a new asymptotic bound on $M(L)$, and no identity $B_* = 1/\mathcal{A}$ is asserted: the support and admissible-object restrictions differ.

6 Formal verification and reproducibility

6.1 Statements and scope

The ancillary Lean development uses Lean 4.19.0 and mathlib at commit

c44e0c8ee63ca166450922a373c7409c5d26b00b

(the recorded v4.19.0 revision). The locked `lake-manifest.json` is included. All numerical parameters are exact rational numbers interpreted in $\mathbb{R}$.

Besides the finite certificate, the development connects the algebra to Lebesgue integrals and proves the six statements in Table 3. The theorem names are in the namespace `Autocorrelation`; the original specification propositions are in `Targets.lean`. The audit file contains explicit type ascriptions binding all six proof terms to those propositions.

Table 3: Formalized statements and their roles in this paper.

Lean theorem	Mathematical content
<code>overlap_integral</code>	The actual correlation integral equals the piecewise expression of Section 2.
<code>witness_admissibility</code>	Nonnegativity, integrability of F_δ and its square, and integrability of every shifted product.
<code>mass_integral</code>	The actual mass integral is $m + \delta/a$.
<code>uniform_correlation</code>	The lower bound $a_{F_\delta}(t) \geq \gamma$ for all $t \in [0, 1]$ and $\delta > 0$.
<code>explicit_main</code>	The ordinary-function statement (1.5), including positivity of the mass.
<code>limiting_main</code>	For every $q < R$, an ordinary-function witness satisfying (1.6), with its admissibility conditions.

The proof of `limiting_main` chooses an actual F_δ , not the atomic measure. It does not assert attainment of R . The real-valued integrals in these statements are accompanied by the necessary integrability proofs. No default value of an integral on a nonintegrable function is used to obtain the bounds.

The formalization does *not* define the supremum $\mathcal{A}$ or formalize the final supremum argument following (1.6); that argument is given in Section 4. Nor does it formalize the applications in Section 5, the Fourier argument, or Appendix A. These exclusions are part of the claimed scope.

6.2 Recorded verification and reproduction

The archived local Windows x64 run used Lean commit 6caae842e94 and Lake 5.0.0-6caae8. It rebuilt all 39 project modules after removal of the project’s build directory, retaining matching dependency caches. The complete lake build, `lake env lean Audit.lean`, and original Windows checking script returned exit code zero. This is a project-source rebuild, not a rebuild of the Lean compiler and all dependencies from source.

The 25 declarations requested by `Audit.lean` report only `propext`, `Classical.choice`, and `Quot.sound`. No `sorryAx`, problem-specific axiom, or external-computation axiom appears in those dependencies. These are proofs relative to Lean’s standard mathematical foundations, not claims of axiom-free verification. The Python programs are not trusted oracles for the Lean proofs.

From the extracted ancillary directory, the independent exact checks are

```
python verify.py certificate.json
python independent_audit.py certificate.json
python verify_applications.py certificate.json
```

The first and third use only the Python standard library. The second uses SymPy; the recorded version is 1.14.0. The input certificate has SHA-256

```
e9b58a1068baa81723e55ad01640260118
ce156ac23c07332cefefc46f552d79.
```

The line break is typographical; the digest is a single 64-character hexadecimal string.

With the pinned Lean toolchain and dependencies installed, proof checking from `anc/lean/` uses

```
lake build
lake env lean Audit.lean
```

The ancillary README gives cache and low-memory build options, source checksums, and the distinction between the original local evidence and subsequent packaging checks. No dependency binaries are bundled. The only recorded mathlib patch concerns the cache unpacker’s worker count; it changes no mathematical declarations.

Rechecking the archived proof sources does not require regeneration. The original generator was byte-identical on the recorded Windows run; it uses native line endings, so its original raw-hash regeneration check is not a cross-platform reproducibility claim. The verified Lean source bytes are preserved in the ancillary archive. During manuscript preparation the exact Python checks were rerun, but Lean was not rerun in the manuscript-preparation environment; the Lean execution claim above refers to the archived local run.

Use of generative AI. ChatGPT and Codex were used to assist with construction and proof exploration, verification code, Lean proof development, and manuscript drafting. The mathematical arguments, exact certificates, and stated formal-verification scope are provided for independent examination; generative output is not itself evidence of correctness. Responsibility for the submitted manuscript rests with its human author or authors.

A A limitation of independent sampling and completion

This appendix gives a separate application of the same mass parameters. It explains why one particular discretization procedure need not preserve a strong continuous bound. All results here are mathematical proofs, not Lean-verified statements. They impose independence only on the initial sampling, not on the subsequent completion algorithm.

A.1 A finite missing-distance estimate

Fix $C \subseteq \{0, \dots, L\}$ and form a random set S by retaining all of C and selecting every other location independently, with probabilities p_i . Set $p_i = 1$ on C , and write

$$u = |C|, \quad \kappa = \mathbb{E}|S|, \quad p_* = \max_{i \notin C} p_i < 1, \quad T = (C - C) \cap \{1, \dots, L\}.$$

Let $n_0 = L - |T| > 0$ and let $H(S)$ count the distances in $\{1, \dots, L\}$ missing from $S - S$.

Proposition A.1. *In this finite model,*

$$\mathbb{E}H(S) \geq n_0 \exp\left(-\frac{\kappa^2 - u^2}{2n_0(1 - p_*)}\right), \quad \text{Var } H(S) \leq \kappa(\kappa^2 + \kappa). \quad (\text{A.1})$$

Proof. For $d \notin T$, put $\lambda_d = \sum_{i=0}^{L-d} p_i p_{i+d}$. The events that a pair $(i, i+d)$ is not selected are decreasing events in the independent underlying bits. Positive association therefore gives

$$\mathbb{P}(d \notin S - S) \geq \prod_{i=0}^{L-d} (1 - p_i p_{i+d}) \geq \exp\left(-\frac{\lambda_d}{1 - p_*}\right). \quad (\text{A.2})$$

For completeness, association for two functions monotone in the same direction follows by induction on the number of bits: conditional covariance is nonnegative by induction, and covariance of the two conditional means for the last bit is $p(1-p)\Delta F \Delta G \geq 0$. Products of decreasing event indicators remain decreasing, so the assertion extends to the intersection in (A.2). The second inequality uses $p_i p_{i+d} \leq p_*$ for $d \notin T$ and $-\log(1-x) \leq x/(1-p_*)$ on $[0, p_*]$.

Every deterministic pair has distance in T . Consequently

$$\sum_{d \notin T} \lambda_d \leq \frac{\kappa^2 - \sum_i p_i^2}{2} - \binom{u}{2} \leq \frac{\kappa^2 - u^2}{2}.$$

Sum (A.2) over the n_0 distances and use convexity of the exponential to obtain the expectation bound.

For the variance, let $H^{(i)}$ be the value after independently resampling bit i , and let M_{-i} count the other selected locations. Changing bit i changes at most M_{-i} distances. Variance tensorization for independent coordinates gives

$$\text{Var } H \leq \frac{1}{2} \sum_i \mathbb{E}(H - H^{(i)})^2 \leq \sum_{i \notin C} p_i(1 - p_i) \mathbb{E}M_{-i}^2 \leq \kappa(\kappa^2 + \kappa).$$

Here $\mathbb{E}M_{-i} \leq \kappa$ and $\text{Var } M_{-i} \leq \kappa$. The tensorization inequality follows by induction from the law of total variance and Jensen's inequality, or equivalently by its conditional-variance form; independent resampling accounts for the factor $1/2$. $\square$

For a fixed set S of size k , any complete set $B \supseteq S$ must satisfy

$$|B|(|B| - 1) \geq k(k - 1) + 2H(S). \quad (\text{A.3})$$

Indeed, adjoining $r = |B \setminus S|$ points introduces at most $kr + \binom{r}{2}$ new unordered pairs, each representing at most one formerly missing positive distance. No restriction on the algorithm choosing those new points is used.

Theorem A.2 (Independent-sampling completion bound). *Suppose a sequence of the preceding models satisfies*

$$\frac{\mathbb{E}|S_L|}{\sqrt{L}} \longrightarrow \bar{m}, \quad \frac{|C_L|}{\sqrt{L}} \longrightarrow \bar{a}, \quad |(C_L - C_L) \cap \{1, \dots, L\}| = o(L), \quad p_* = o(1).$$

Put $\beta = \exp[-(\bar{m}^2 - \bar{a}^2)/2]$. For each fixed $\varepsilon > 0$, with probability tending to one,

$$\frac{H(S_L)}{L} \geq \beta - \varepsilon, \quad \frac{|B|}{\sqrt{L}} \geq \sqrt{\bar{m}^2 + 2\beta} - \varepsilon \quad \text{for every complete } B \supseteq S_L. \quad (\text{A.4})$$

Proof. Proposition A.1 gives $\liminf \mathbb{E}H(S_L)/L \geq \beta$ and $\text{Var } H(S_L) = O(L^{3/2})$. Hence $(H - \mathbb{E}H)/L \rightarrow 0$ in probability. Also $\text{Var } |S_L| \leq \mathbb{E}|S_L| = O(\sqrt{L})$, so $|S_L|/\sqrt{L} \rightarrow \bar{m}$ in probability. Apply (A.3) on the intersection of these high-probability events. For example, its positive quadratic root gives $|B| \geq (1 + \sqrt{1 + 4k(k-1)} + 8\bar{H})/2$ simultaneously for all completions, and division by $\sqrt{L}$ gives the second assertion. $\square$

A.2 The parameters of the polynomial construction

For the normalized measure $\mu/\sqrt{\gamma}$, let

$$\bar{a} = \frac{a}{\sqrt{\gamma}}, \quad \bar{m} = \frac{m}{\sqrt{\gamma}}.$$

Choose $C_L = \{0, \dots, \lfloor \bar{a}\sqrt{L} \rfloor - 1\}$ and, outside C_L , select location i independently with probability

$$p_i = \frac{h(i/L)}{\sqrt{\gamma}\sqrt{L}}.$$

For sufficiently large L these are valid probabilities, the short initial cluster is disjoint from the support of $h(i/L)$, and Riemann sums give the mass limits of Theorem A.2. Moreover $p_* = O(L^{-1/2})$ and the cluster represents only $O(\sqrt{L})$ distinct positive distances. The theorem therefore applies with

$$\begin{aligned} \beta_0 &= \exp\left(-\frac{m^2 - a^2}{2\gamma}\right) = 0.3598474622246835\dots, \\ R_0 &= \sqrt{\frac{m^2}{\gamma} + 2\beta_0} = 1.7766465203957698\dots > 1.7766 > \sqrt{3}. \end{aligned} \tag{A.5}$$

In particular, with probability tending to one every completion requires at least $(1.7766 - o(1))\sqrt{L}$ points, understood in the fixed-tolerance sense of (A.4). The conclusion concerns this initial sampling and retention of its points, not all random constructions or all complete rulers.

The same framework has a coarse obstruction independent of the particular coefficients. If $\bar{m}$ is the mass of any finite positive measure ν satisfying $\nu * \tilde{\nu} \geq \mathbf{1}_{[-1,1]} dt$, the Fourier estimate at $\xi = 3\pi/2$ gives

$$\bar{m}^2 \geq 2 + \frac{4}{3\pi} > \frac{80}{33}.$$

Since $z + 2e^{-z/2}$ is strictly increasing for $z > 0$,

$$\bar{m}^2 + 2e^{-(\bar{m}^2 - \bar{a}^2)/2} \geq \bar{m}^2 + 2e^{-\bar{m}^2/2} > \frac{80}{33} + 2e^{-40/33} > 3. \tag{A.6}$$

Thus improving only the continuous density cannot overcome the cost within this specific independent-sampling-and-retention framework. Correlated initial constructions, conditioning on rare samples, or replacing a positive proportion of the sampled points are not ruled out.

The strict numerical inequalities in (A.5) and (A.6) are checked by `verify_applications.py` with rational arithmetic. For rational $x \geq 0$, the elementary estimate

$$e^x \leq \sum_{j=0}^J \frac{x^j}{j!} + \frac{x^{J+1}/(J+1)!}{1 - x/(J+2)} \quad \left(\frac{x}{J+2} < 1\right)$$

provides, on taking reciprocals, a rational lower bound for e^{-x} . We use $J = 30$ and compare squares to avoid using rounded decimal square roots as proofs.

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